

Municipal Revenue Intelligence & Analytics

Project Description & Objectives

1. Project Overview

Municipal Revenue Intelligence & Analytics is a data analytics project designed to demonstrate how municipal billing and payment data can be transformed into useful financial and operational insights. The project uses a synthetic municipal revenue dataset and follows a practical analytics workflow from data generation and exploration through SQL analysis and dashboard visualization.

2. The Business Problem

Municipalities generate large volumes of billing and payment information across customers, services, municipal areas and time periods. Raw transaction-level data does not immediately show where revenue is being generated, where collections are weaker, where outstanding balances are accumulating, or which customer segments require closer attention.

The project therefore aims to turn municipal revenue data into a clear decision-support view of billing, collections, outstanding balances and collection risk.

3. What We Are Trying to Achieve

- **Measure financial performance:** understand total billed revenue, total collections and outstanding balances.
- **Compare municipalities:** identify differences in billing, collections, outstanding balances and collection performance.
- **Understand services:** analyse revenue and outstanding balances across Electricity, Water, Waste and Property Rates.
- **Analyse payment behaviour:** examine Paid, Partial and Unpaid billing records.
- **Identify collection risk:** highlight overdue balances and customers with significant outstanding amounts.
- **Understand trends:** monitor how billing, collections and outstanding balances change over time.
- **Build a decision-ready dashboard:** present the results in an accessible visual format for business users.

4. Dataset & Synthetic Data

The project uses a synthetic dataset generated in Databricks so that the complete data-generation process is reproducible and can represent a realistic municipal revenue use case without relying on confidential customer information.

Dataset characteristic	Project setup
Customers	25,000 synthetic customers

Dataset characteristic	Project setup
Period	January 2024 – December 2025
Billing records	Approximately 600,000 records
Provinces	Gauteng, Limpopo, Mpumalanga, KwaZulu-Natal
Services	Electricity, Water, Waste, Property Rates
Customer types	Residential, Business
Payment status	Paid, Partial, Unpaid
Payment methods	EFT, Card, Cash, Debit Order, None

5. Analytics Architecture

Stage	Technology	Purpose
1. Synthetic Data Generation	Databricks	Generate realistic customers, billing, payments and balances.
2. Data Exploration	Python	Assess structure, distributions, quality and patterns.
3. Analytical Queries	Databricks SQL	Calculate financial, geographic, service and risk metrics.
4. Dashboard	Data Studio / Looker Studio	Communicate insights through interactive visualizations.

6. Core Analytical Questions

- How much revenue was billed and how much was collected?
- How much revenue remains outstanding?
- How does collection performance differ between municipalities?
- Which services generate the largest amounts of billed revenue and outstanding balances?
- What proportion of billing is Paid, Partial or Unpaid?
- Where are overdue balances concentrated?
- Which customer segments contribute most to outstanding balances?
- Which accounts meet the project's high-risk criteria?
- How do billing, collections and outstanding balances change over time?

7. Dashboard Goal

The final dashboard should function as a Municipal Revenue Intelligence view rather than simply a collection of charts. A user should be able to move from an overall financial snapshot to municipality performance, service performance, collection behaviour, trends and revenue risk.

Dashboard area	What it should show
Financial Overview	Total billed, total collected, total outstanding, collection rate and customer/record counts.
Municipality Performance	Billed, collected, outstanding, collection rate and overdue balances by municipality.
Service Performance	Revenue and outstanding balances across municipal services.

Dashboard area	What it should show
Collection Behaviour	Paid, Partial and Unpaid activity and payment patterns.
Trends	Monthly movement in billing, collections and outstanding balances.
Debt & Risk	Overdue balances, days overdue and high-risk customer accounts.

8. Key Derived Metrics

Some metrics are not stored as original transaction columns and therefore need to be calculated in the analytics layer or dashboard layer. The project should deliberately distinguish raw fields from derived metrics.

Metric	Definition / purpose
Collection Rate	Total collected ÷ total billed × 100.
Outstanding Balance	Billing amount – payment amount.
Overdue Accounts	Accounts/records with days_overdue > 0.
High-Risk Account	Project rule: total outstanding > 5,000 and maximum days overdue > 90.
Average Days Overdue	Average number of overdue days for relevant records.

9. Expected Project Outcome

The final deliverable will demonstrate an end-to-end analytics workflow: generating a realistic dataset, examining data with Python, performing business-focused analysis with SQL, creating dashboard-ready metrics, and communicating the results through an interactive visualization layer.

The portfolio value: the project demonstrates practical skills in data engineering, exploratory data analysis, SQL, financial/business analytics, metric design and data visualization within one coherent use case.

10. Project Success Criteria

- The synthetic dataset is reproducible and logically consistent.
- The cleaned data is suitable for analysis and visualization.
- Python EDA documents the structure, quality and important patterns in the data.
- SQL produces reliable business metrics and analytical outputs.
- Dashboard calculations are clearly defined and reproducible.
- The final dashboard answers the core business questions without requiring the user to inspect raw data.

Project focus: Municipal Revenue Intelligence & Analytics | Data generated in Databricks | Analysis with Python and SQL | Visualization with Data Studio / Looker Studio